OSF Supplementary Material Power Analyses

This document provides a more detailed explanation of additional power analyses for the manuscript "Investigating letter confusability from visual similarity across orthographic contexts".

We had predicted that low-level effects of visual feature similarity would be stronger where there is less higher-level orthographic information available. However, the interaction between orthographic context and visual feature similarity was not significant. Our interpretation of this null effect is limited, as despite having a relatively large sample size (N=72) and a repeated-measures design, our study was probably underpowered to detect this interaction (Brysbaert, 2019). It was not appropriate to conduct power analyses on the observed data, as post-hoc power analyses on null findings violate two main assumptions (Dziak et al., 2018). Firstly, statistical power reflects the probability of rejecting a null hypothesis, based on estimates of parameters (e.g. effect sizes and variance) at population level (Cohen, 1992). However, power analyses do not predict the probability of a future outcome if they are performed on a test that has already been conducted on existing data. Secondly, power estimates are calculated using sampling distribution, which forecasts probability over a range of possible samples. Post-hoc power analyses are based on a single specific sample rather than a sampling distribution. Consequently, post-hoc power estimates can be misleading, and provide redundant information at best. Dziak et al. (2018) convey this with a quote by Goldstein (1964, p.62): "If the axe chopped down the tree, it was sharp enough". Instead, we simulated new datasets based on the parameters of our original experiment. This enabled us to run a series on power simulations across a range of effect sizes and sample sizes. These analyses allowed us to estimate the effect sizes that we had adequate power to reliably detect with our sample size, as well as the sample sizes required to detect alternative effect sizes of varying magnitude.

Effect size required to be reliably detected with the sample size used in our study

We ran Monte Carlo power analyses on simulated datasets to estimate the interaction effect sizes that could have been reliably detected with our sample size. Power analyses were conducted using the *simr* package (Version 1.0.5; Green & MacLeod, 2016) in *R* (Version 4.0.4; R Core Team 2016). We systematically increased hypothetical interaction effect sizes by $\beta = 0.05$ and ran 50 simulations for each increment, beginning at $\beta = 0.1$. For each simulation, we modelled a larger effect between words and consonant strings relative to words and pseudowords under our hypothesis that visual similarity effects would have a greater impact on letter identification when less orthographic information is available. The simulated power estimations for incremental increases in effect sizes are displayed in Table S1.

	Words vs. Consonant Strings (simulated β value)								
Words vs. Pseudowords (simulated β value)	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
0.10	16%	22%	32%	42%	62%	74%	84%	90%	96%
0.15		28%	38%	52%	58%	64%	82%	90%	94%
0.20			44%	52%	62%	68%	78%	90%	92%
0.25				60%	66%	72%	78%	88%	92%
0.30					68%	74%	80%	90%	94%
0.35						80%	86%	92%	95%
0.40							90%	92%	94%
0.45								96%	98%
0.50									96%

Table S1. Monte-Carlo power simulations (50 simulations per effect size comparison) estimating the interaction effect size that we would have been reliably able to detect with our sample size (N=72). Yellow squares display power estimates of effect sizes (β) *predicted under our hypothesis that visual similarity effects will be greater when less orthographic information is available (consonant strings > pseudowords > words). Orange squares display power estimates of effect sizes predicted under an alternative hypothesis that similarity effects will be greater when the letter string does not correspond with a known word (consonant strings = pseudowords > words).*

Our sample size (N=72) yielded 80% power to detect an interaction with an effect size of β = 0.3 between visual similarity differences in words and pseudowords, and an effect size of β = 0.4 between visual similarity differences in words and consonant strings. These are typical simple effect sizes in psychology, but large interaction effect sizes. The equivalent odds ratios demonstrate that we had the power to detect an interaction if the benefit of having two visually distinct letters was at least 1.35 times more likely to improve letter discrimination in pseudowords relative to words, and 1.49 times more likely in consonant strings relative to words. These analyses suggest that, if there was an undetected interaction between visual similarity and orthographic context in our data, it was smaller than the effect sizes stated above.

Sample size required to detect interaction effect sizes observed in our study

As a secondary measure, we simulated new datasets and ran Monte Carlo power simulations over a range of sample sizes to estimate the sample size required to reliably detect an interaction between visual feature similarity and orthographic status, based on the effect sizes observed in our data. We note that this method is less reliable than the method above, as the reported non-significant effect sizes are unreliable estimates in an under-powered study. Therefore, this power analysis is only included in supplementary materials to demonstrate how else we explored the data.

Again, data simulations and power analyses were conducted using the *simr* package (Version 1.0.5; Green & MacLeod, 2016) in *R* (Version 4.0.4; R Core Team 2016). First, we simulated a dataset based on the parameters and values in our observed data, before entering the simulated data into the maximal model defined previously. Next, we redefined the β values (the logit transformed fixed effect coefficient, which refers to the model estimated effect size) according to the β values observed in our experiment data. A post-hoc analysis of the maximal model (which included the interaction term) revealed that effect sizes for non-significant interactions between visual feature similarity and orthographic status were very small ($\beta < 0.065$). The benefit of having two visually distinct letters was 1.06 times more likely to improve letter discrimination in pseudowords relative to consonant strings, $\beta=0.06$, $OR=1.06$, $SE=0.10$, $Z=0.57$, $p=.569$, and 1.01 times more likely in words relative to consonant strings, $\beta=0.01$, $OR=1.01$, $SE=0.12$, $Z=0.57$, $p=.946$.

We then estimated power based on Monte Carlo simulations over a range of sample sizes. Simulated sample sizes ranged from 200 to 2000, and we ran 50 simulations for each increment in sample size (Figure S1, Appendix A). The power simulations demonstrated that a sample size of 2000 participants would only yield 50% power based on the interaction effect sizes observed in our experiment.

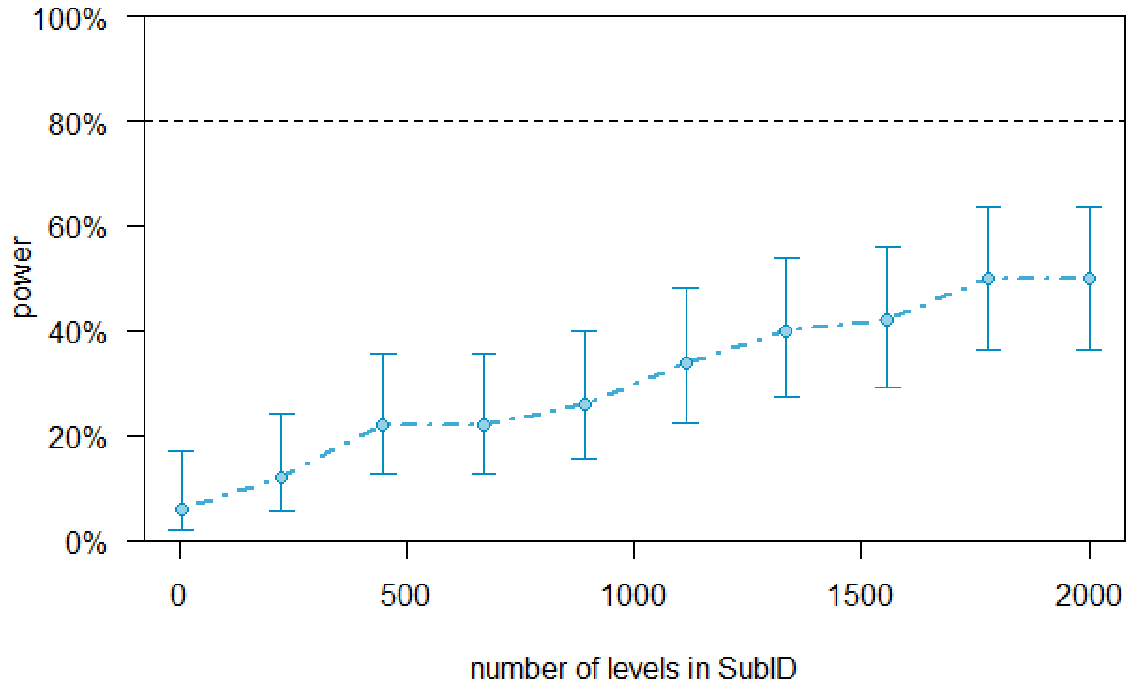

Figure S1. Monte-Carlo power simulations (50 simulations per sample size) predicting how well powered a future study would be to test for an interaction between orthographic status and visual overlap across different sample sizes up to 2000. Effect sizes are based on our original experiment.

We then extended the sample size of the simulated dataset to 10000 participants and repeated the steps above (Figure S2, Appendix B). These power simulations revealed that we would require an approximate sample size of 5000 participants in order to detect an interaction of this size, based on the effect sizes observed within our experiment and 80% power. This does not necessarily entail that a significant interaction would be observed within a larger sample. The increase power would enable researchers to confirm or reject the null hypothesis based on effect sizes of this size with greater certainty.

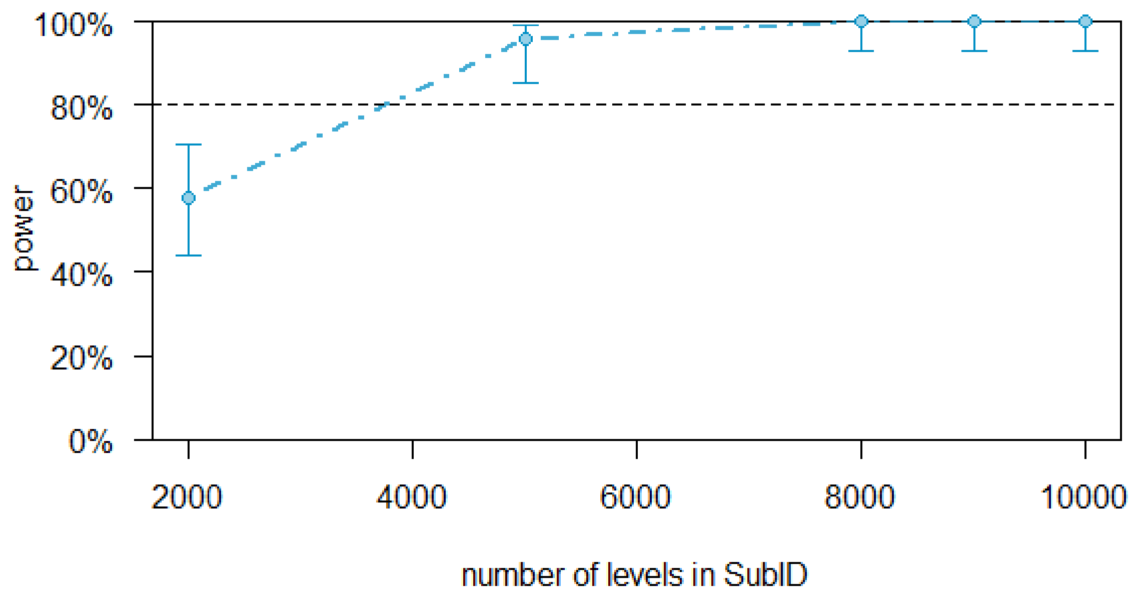

Figure S2. Monte-Carlo power simulations (50 simulations per sample size) predicting how well powered a future study would be to test for an interaction between orthographic status and visual overlap across different sample sizes up to 10000. Effect sizes are based on our original experiment.

References

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Appendix A – estimated power for simulated sample sizes based on effect sizes observed with our experiment (up to 2000 participants)

Power for model comparison, (95% confidence interval), by number of levels in SubID:

3: 6.00% (1.25, 16.55) - 432 rows

225: 12.00% (4.53, 24.31) - 32400 rows

447: 22.00% (11.53, 35.96) - 64368 rows

669: 22.00% (11.53, 35.96) - 96336 rows

891: 26.00% (14.63, 40.34) - 128304 rows

1113: 34.00% (21.21, 48.77) - 160272 rows

1335: 40.00% (26.41, 54.82) - 192240 rows

1557: 42.00% (28.19, 56.79) - 224208 rows

1779: 50.00% (35.53, 64.47) - 256176 rows

2001: 50.00% (35.53, 64.47) - 288144 rows

Appendix B – estimated power for simulated sample sizes based on effect sizes observed with our experiment (up to 10000 participants)

Power for model comparison, (95% confidence interval), by number of levels in SubID:

2000: 58.00% (43.21, 71.81) - 288000 rows

5000: 96.00% (86.29, 99.51) - 720000 rows

8000: 100.0% (92.89, 100.0) - 1152000 rows

9000: 100.0% (92.89, 100.0) - 1296000 rows

10000: 100.0% (92.89, 100.0) - 1440000 rows